Tutorial Questions 03 – Functions Dr Killian O'Brien
6G4Z9019 Mathematical Methods 04/11/26

When possible you should work on the questions in advance and be prepared for discussions on them with your colleagues and tutor. Some questions are straightforward calculation while others require more discussion and thought. The computer icon 💻 indicates where computer calculation, plotting or programming may be of use.

Further questions can be found in the lecture notes as well as the recommended unit resources (see Moodle).

Domains and co-domains

1. 💻 Consider a function $\phi : X \to Y$, where X, Y are subsets of $\mathbb{R}$, defined by the formula

$$\phi(x) = \frac{x^2 + 1}{2x - 3}.$$

Can you choose the subsets X and Y so that the function ϕ is variously

a) injective and surjective, (i.e. bijective)

b) surjective but not injective,

c) injective but not surjective,

d) neither injective nor surjective.

If X and Y are both the empty set, i.e. $X = Y = \{\}$, is ϕ injective and/or surjective?

Solution:
One way to approach this is to view a plot of the graph of ϕ. There is a singularity at $x = 3/2$ and plotting it either side of that indicates the general behaviour

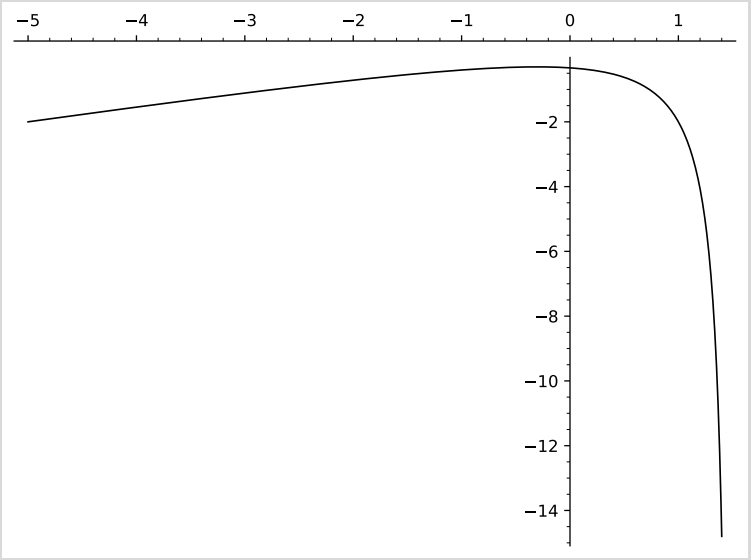


Fig: The graph of ϕ to the left of $3/2$.

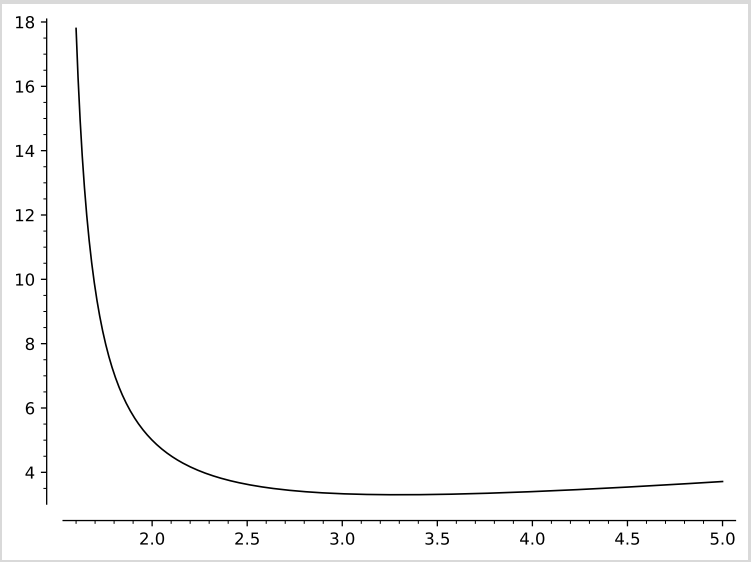


Fig: The graph of ϕ to the right of $3/2$.

As well as the singularity at $x = 3/2$ there appear to be two turning points to either side of this as well. The plots show that there is an interval of values surrounding 0 which do not occur as values of the function. We can identify these by consider when solutions exist to

$$y = f(x) = \frac{x^2 + 1}{2x - 3}.$$

This is equivalent to

$$x^2 - 2xy + 1 + 3y = 0.$$

Considering this as a quadratic in x it will fail to have any solution(s) if and only if its discriminant is negative, i.e.

$$4y^2 - 4(1 + 3y) < 0$$

which is equivalent to

$$y^2 - 3y - 1 < 0.$$

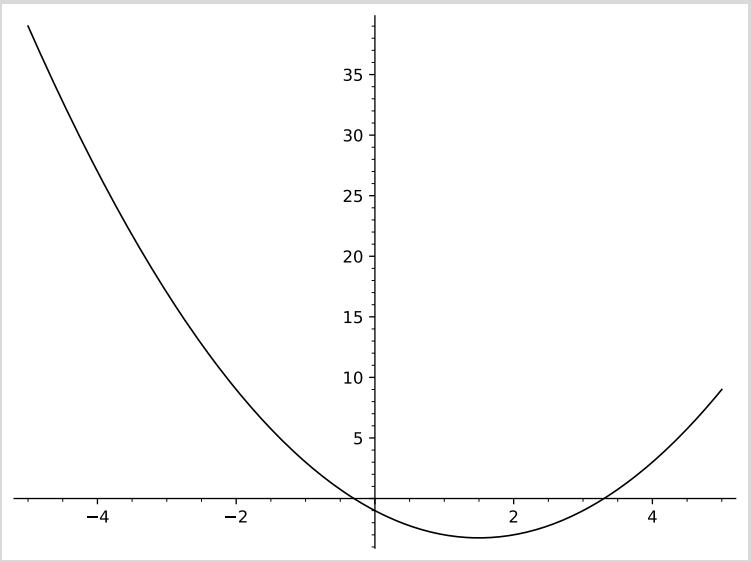


Fig: The plot of $y^2 - 3y - 1$

From the plot we see that $y^2 - 3y - 1$ is negative between its two zeroes which are at

$$y = 3/2 \pm \frac{1}{2}\sqrt{13} \approx 3.303, -0.303.$$

So the interval between these values can inform our choices for the co-domain concerning surjectivity and the location of the turning points of ϕ which are at the zeroes of $\frac{d\phi}{dx} = \frac{2\left(x^2-3\,x-1\right)}{(2\,x-3)^2}$ which are also $3/2 \pm \frac{1}{2}\sqrt{13}$, these can inform our choices of domain concerning injectivity.
So some choices for $\phi : X \to Y$ are

a) injective and surjective, (i.e. bijective)

$$X = (-4, -3), \quad Y = (\phi(-4), \phi(-3)) = (-17/11, -10/9)$$

b) surjective but not injective, (we simply enlarge the domain slightly to include the turning point and some points beyond it, note that maximum value achieved at the turning point is $y_0 = 3/2 - \frac{1}{2}\sqrt{13} \approx -0.303$,

$$X = (-4, 0), \quad Y = (\phi(-4), y_0].$$

c) injective but not surjective, (take our injective & surjective example above and extend the co-domain to destroy the surjectivity)

$$X = (-4, -3), \quad Y = \mathbb{R}.$$

d) neither injective nor surjective.

$$X = \mathbb{R} \backslash \left\{\frac{3}{2}\right\}, \quad Y = \mathbb{R}.$$

2. For a function $\psi : X \to Y$ defined by $\psi(x) = \tan(x)$ answer parts (a) - (d) of question 1.

Solution:
Relevant facts about $\tan(x)$ are that it is periodic with period π, it has singularities at every odd multiple of $\pi/2$ and over the range $\pi/2 < x < \pi/2$ it takes on every value from $\mathbb{R}$.

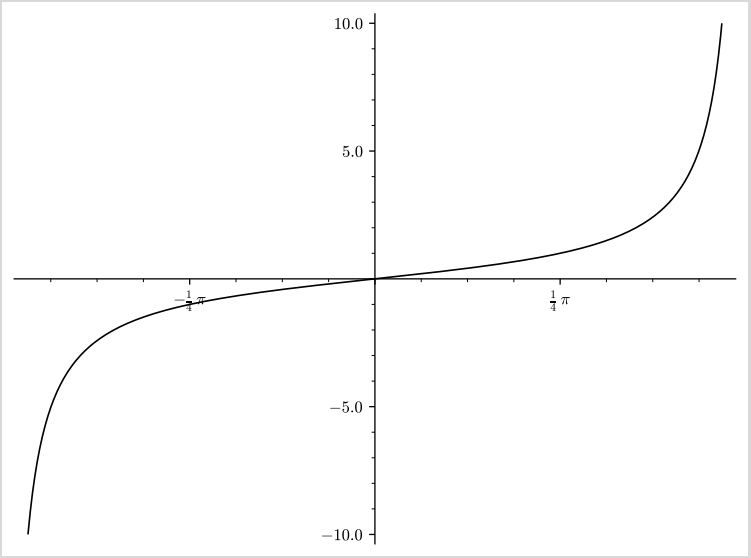


Fig: The plot of $\tan(x)$

With these facts in mind we give example domains and co-domains

a) injective and surjective, (i.e. bijective)

$$X = (-\pi/2, \pi/2), \quad Y = \mathbb{R},$$

b) surjective but not injective,

$$X = (-\pi/2, \pi/2) \cup (\pi/2, 3\pi/2), \quad Y = \mathbb{R},$$

c) injective but not surjective,

$$X = (0, \pi/2), \quad Y = \mathbb{R},$$

d) neither injective nor surjective,

$$X = (0, \pi/2) \cup (\pi, 3\pi/2), \quad Y = \mathbb{R}.$$

Composition of functions and inverses

3. Consider the functions f, g, h, all $\mathbb{R} \to \mathbb{R}$, defined by the formulas

$$f(x) = -2x - 3, \quad g(x) = x^2 + 3x + 1, \quad h(x) = x/2.$$

Find the formulas, simplified as far as possible, that represent the functions $f \circ g \circ h$, $g \circ f \circ h$, $h \circ g \circ f$, $f \circ f$, $g \circ g$ and $h \circ h$.

Solution:

$$\begin{aligned}
(f \circ g \circ h)(x) &= f(g(h(x))) = -\frac{1}{2}\,x^2 - 3\,x - 5 \\
(g \circ f \circ h)(x) &= g(f(h(x))) = x^2 + 3\,x + 1 \\
(h \circ g \circ f)(x) &= h(g(f(x))) = 2\,x^2 + 3\,x + \frac{1}{2} \\
(f \circ f)(x) &= f(f(x)) = 4\,x + 3 \\
(g \circ g)(x) &= g(g(x)) = x^4 + 6\,x^3 + 14\,x^2 + 15\,x + 5 \\
(h \circ h)(x) &= h(h(x)) = \frac{1}{4}\,x
\end{aligned}$$

4. Can you compose your own relatively simple, but not trivial, examples to show that composition of functions is not commutative, i.e. that in general $f \circ g \neq g \circ f$?

5. Consider the functions $f : (0, \infty) \to (0, \infty)$ and $g : (0, \infty) \to \mathbb{R}$ defined by

$$f(x) = x^2, \quad g(x) = \log(x).$$

Find the definitions of the functions $g \circ f$ and $(g \circ f)^{-1}$ and confirm in this case that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Solution:

$$
\begin{aligned}
(g \circ f)(x) &= g(f(x)) = \log\left(x^2\right) = 2\log(x) \\
(g \circ f)^{-1}(x) &= e^{x/2} \\
f^{-1}(x) &= x^{1/2} \\
g^{-1}(x) &= e^{x} \\
(f^{-1} \circ g^{-1})(x) &= f^{-1}(g^{-1}(x)) = (e^{x})^{1/2} = e^{x/2}
\end{aligned}
$$

Partial fractions

6. Obtain the partial fraction expansions of the following rational function expressions

a)

$$-\frac{13\,t-23}{2\,(3\,t-1)(t-3)},$$

Solution:

$$-\frac{7}{2\,(3\,t-1)}-\frac{1}{t-3}$$

b)

$$-\frac{9\,t^2+10\,t+15}{3\,(t^2+1)(3\,t-5)},$$

Solution:

$$\frac{2\,t}{3\,(t^2+1)}-\frac{5}{3\,t-5}$$

c)

$$\frac{2\,t^2-11\,t+14}{(5\,t-4)(t+1)^2},$$

Solution:

$$\frac{2}{5\,t-4}-\frac{3}{(t+1)^2}$$